Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Lets go
Selected Domain	Domain 2 - ALPHA RED / Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8 - Decent Work and Economic Growth
Project / Solution Name	Milestone2Cash Proof-to-Cash
Name (Leader)	Jiazhong Yang
Email & Phone (Leader)	yjzae41475@gmail.com +60 173072181
University Affiliation(s)	Monash University Malaysia; Universiti Malaya
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Owners and finance staff in Malaysian project-based service firms handle milestone records, invoices, customer replies, transfer screenshots and bank statements separately. Re-entering and comparing these records can hide unpaid balances or lead to reminders for the wrong amount.

Experian reported SME payment delays of 64 days in early 2025, illustrating wider cash-flow pressure [1]. Our synthetic scenario is an overdue RM18,000 invoice receiving RM9,000. The remaining RM9,000 must stay collectible. A promise affects expected timing without becoming cash. Milestone2Cash connects the evidence, human decision and balance. Customer interviews and measured business savings are not claimed.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

A completed milestone creates an invoice draft for Owner approval. Finance reviews a reminder and confirms simulated sending, then pastes a customer reply or uploads a screenshot.

Tesseract reads the image locally; DeepSeek structures protected text. Uncertain or invalid results go to review. Rules compare reviewed fields with imported bank credits and identify partial matches, mismatches and duplicates. Finance approves an eligible match before SQLite records payment, balance and audit changes.

Paying RM9,000 against RM18,000 leaves RM9,000 collectible. A confirmed promise changes expected timing without creating cash. The workflow combines GenAI, database and Replit hosting skills, with role controls and reuse across two synthetic companies. Appendix A describes Orbyte interface assumptions. No live Orbyte connection is claimed.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

React/TypeScript calls seventeen Node.js API operations backed by SQLite. Local Tesseract OCR and DeepSeek deepseek-flash support protected-text extraction, reply interpretation and reminder drafting. Images and full bank CSVs are not sent to the model.

Rules handle matching and forecasts. Server checks enforce company separation, demo roles, evidence versions and duplicate-allocation protection. Manual correction is available. Appendix A describes the connected blocks.

Sponsor cloud: Replit hosts the application. DeepSeek uses an existing team API account, not a claimed sponsor service. RunPod and live Orbyte are not used. API Premium compliance and hosted-API RunPod applicability await organiser clarification. Hosted SQLite is resettable demo storage.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We mapped one service-business journey around its unpaid balance, evidence and human decisions. The interface shows original receipts, extracted fields and matching reasons; English, Chinese and Malay views support learning. Shared contracts and synthetic tests guide implementation.

The Replit prototype is accessible; redeployment can reset data. Phase 2 will test new receipt layouts, run a consented manual-versus-tool workflow study, and add durable storage and production authentication. Ambiguous evidence and AI errors are the main risks, addressed through review and manual correction. No customer interviews, measured staff-time savings or independent team review are claimed. Appendix A records current technical validation.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8.2, productivity through technological upgrading and innovation. Secondary: SDG 8.3, support for entrepreneurship and MSME growth. More reliable record handling is the intended contribution; UN economic indicators are not measured by this prototype. [2]
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published context: SME payment delay averaged 64 days in early 2025 (Experian), not a product outcome [1]. Technical baseline: 35/40 receipt references were exact, or 87.5%, on the frozen synthetic evaluation set [3]. Staff-time baseline has not been measured.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Reference-field exact-match rate (% of all held-out receipt cases). Keep rejected outputs in the denominator. Track unsafe match candidates separately. In a future consented pilot, record minutes per correctly verified invoice and correction count using the same evidence for manual and tool workflows.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed technical target: at least 95% exact references on a new held-out synthetic set, up 7.5 percentage points from 87.5%, while retaining zero unsafe match candidates. This is a correction-work proxy, not measured productivity. Staff-time saving targets will follow a measured baseline.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

The leader should review the completed document before ticking the final checklist item.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

The team confirms no self-owned code from before the competition was used. Earlier iterations during this same competition are part of this event's development history.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js 24; SQLite	Runtime / database code	Node.js licence; SQLite public domain
React; Vite; Vitest; Ajv	Interface / build / test code	MIT
TypeScript; Playwright; Tesseract.js; sharp	Code	Apache-2.0; libvips LGPL notices
Tesseract eng / chi_sim data	Pretrained OCR data	Package MIT and model notices
DeepSeek deepseek-flash	Hosted model API	Provider API terms

Full versions, datasets, development AI assistance and service terms are listed in the Technical and Tools Declaration. Project-created synthetic data is not third-party customer data.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

The team confirms no employment or internship intellectual-property restrictions.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Team leader: Jiazhong Yang

Signature: Jiazhong Yang

Date: 25 October 2026

Appendix A Architecture and evidence

System blocks

1. React sends company-scoped requests to the Node.js API.

2. Local OCR reads server-held images. DeepSeek receives protected text; uncertain meaning requires review.

3. Finance reviews evidence. Rules compare it with imported bank credits, including amount, reference, timing and duplicates.

4. Finance approval rechecks evidence; one SQLite transaction records allocation, payment, balance and audit changes.

5. Forecasts keep confirmed promises separate from actual cash. Promises create no payment.

Orbyte reuse: declared customer, project, invoice and evidence interfaces would return review tasks, balances and audit events. This proposed module does not have a live Orbyte connection.

Evidence and limitations

The recorded 24 October evaluation used 40 synthetic receipts and 40 replies: amount/time 40/40, reference 35/40 and intent 39/40. Matching passed 30 cases; 40 daily forecast comparisons matched hand calculations [3]. These model scores predate the latest reply-semantic safeguards; the 40-item provider benchmark has not been rerun. Synthetic checks do not establish real-bank accuracy or business impact.

Application checks on 24 October 2026: 60 backend tests, 23 frontend tests, 13 contract/data groups and a successful build at c438678. Hosted SQLite data can reset on redeployment. Demo identity selection, messaging and business records are synthetic. No real bank, WhatsApp, tax or Orbyte service is connected.

On 25 October 2026 (01:30-01:50 MYT), Codex browser actions passed F01-F06 on the public Replit app: role approvals, edited AI reminder/simulated sending, partial/final payments, promise timing, duplicates and company separation. Two receipts and one promise showed 95% AI confidence without manual field correction. RM18,000 settled; a separate RM6,000 invoice remained unpaid. The promise changed expected timing, not cash or the original due date. The 390-pixel viewport passed inspected EN/Chinese/Malay and both-theme checks. This was not an independent human/SME, time-saving, physical-phone or redeployment-persistence study. Evidence: submission/validation/results.json and screenshots.

References and access

[1] Experian State of Credit 2025; [2] UN Goal 8 targets; [3] STEP-7 evaluation; Live prototype; Source repository; Midday design reference

Midday supplied design references only; no source code or assets were copied.